SIME DARBY PROPERTY UNIVERSITY HACKATHON 2026 · PLANET TRACK

CircularFit

Don’t skip it. Share it.

Routing handover renovation waste to a neighbour, a salvage partner or a recycler, before landfill.



Team name	Team leader	University	Members (3 to 5)	Email and phone
[fill in]	[fill in]	[fill in]	[fill in]	[fill in]

1 · THE PROBLEM

Every handover starts a renovation rush

- Hundreds of homes are handed over in one precinct
- Owners strip out sinks, doors, cabinets and spare tiles, often unused
- It all goes into skip bins, then landfill
- Landfill space in Malaysia is already tight

~16% of materials used in housing becomes waste

~32% of that waste could be reused or recycled

8.8 Mt of housing waste a year in Malaysia (estimate)

Sources: Ain Shams Engineering Journal (2021); UTHM study on construction waste and landfill space

2 · TARGET USERS

Three people, one shared problem

Resident or contractor

Wants a quick, free way to clear strip-out without extra skip bins.

GETS

A one-minute log and green points

Neighbour

Needs a few tiles, a door or a sink, and would rather not buy new.

GETS

Free materials close to home

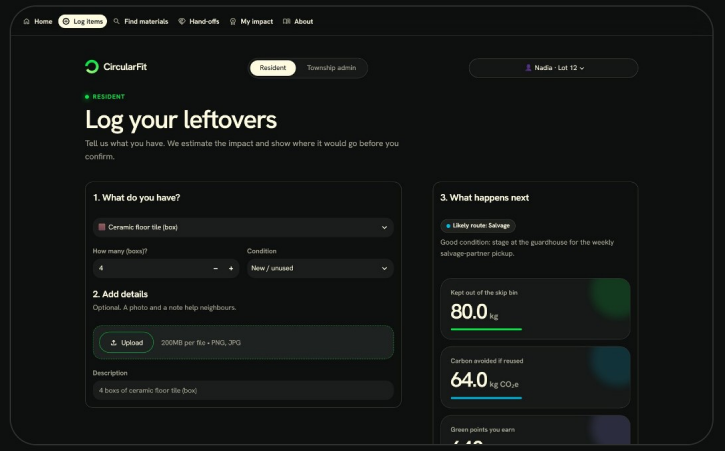
Township management

Needs less waste, sustainability numbers and a plan for the next handover wave.

GETS

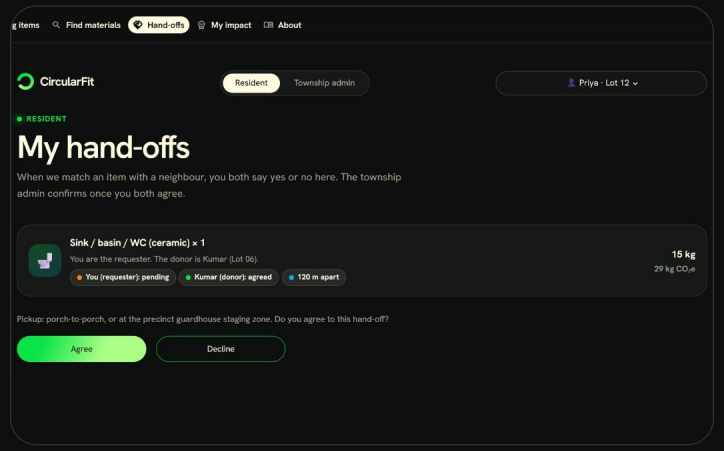
A planner, a routing console and a ledger

Log it. Route it. Prove it.



Log

A resident logs leftovers and sees the likely route first.



Agree

Donor and requester each say yes or no on their own page.



Prove

The admin confirms, and the ledger feeds the dashboard.

Screens from the working prototype (synthetic data)

4 · INNOVATION

Not another marketplace

What changes	Today	With CircularFit
Where strip-out goes	Skip bin, then landfill	Best route first, landfill last
Who sees the numbers	Nobody	Township dashboard and ledger
Planning the next wave	Guesswork	Handover planner
Resident incentive	None	Green points
Proof it happened	None	Two-sided agreement, admin confirms

Waste-hierarchy routing
Decided per item, not per listing

Hyper-local
Matched within a few hundred metres

Handover planner
Forecasts waste for a whole precinct

Auditable ledger
Only confirmed hand-offs count

5 · HOW IT WORKS

Six steps from strip-out to ledger

1. **Log**
   Material, quantity, condition
2. **Estimate**
   Weight and carbon from a catalogue
3. **Route**
   The first route that applies wins
4. **Agree**
   Donor and requester say yes or no
5. **Confirm**
   Admin confirms after both agree
6. **Report**
   Ledger, dashboard and points

Reuse
A neighbour asked for it

Salvage
Staged for weekly pickup

Recycle
Collected by a recycler

Landfill
Only as a last resort

Reuse match score: 40% material · 25% distance · 20% wording · 15% quantity

One handover wave, tens of tonnes diverted

Planning scenario: 300 homes · 60% renovating · 900 kg each · 30% of divertible waste captured

162
tonnes of renovation waste expected

52
tonnes divertible (32%)

5.2
of 54 skip-bin loads avoided

10.9
tonnes CO_2e avoided (estimate)

Township: less waste to haul, numbers to report

Residents: free materials and green points

Environment: less landfill, less new material

All inputs are assumptions. Replace waste per home with precinct data in a pilot.

7 · FEASIBILITY

It already runs

What exists today

- Working prototype in Python and Streamlit
- Full loop: log, route, agree, confirm, report
- No proprietary data needed
- Carbon factors from public databases
- Needs guardhouse space, a salvage partner, a recycler and JMB support

Risk	How we handle it
Low adoption	Launch at handover, add green points
Contractors prefer skip bins	Quick batch listing for a whole strip-out
Safety of used fixtures	Condition check and a no-warranty notice
Batches too small to collect	Staging zone and a recycler minimum

8 · IMPLEMENTATION

Pilot one handover wave, then scale

PHASE 0 · WEEKS 0–4

Agree and baseline

- Pick the pilot precinct
- Agree staging space and partners
- Measure waste per home today

PHASE 1 · WEEKS 5–16

Run one wave

- Resident and admin roles live
- Track tonnes, bins and carbon
- Tune routing rules and rewards

PHASE 2 · NEXT

Township rollout

- Real sign-in from handover records
- Reminders and QR check-in
- Regular partner pickups

PHASE 3 · LATER

Beyond one township

- Other developers
- Townships across Malaysia
- A shared partner network

Developer: sponsor, staging space

JMB: resident access

Partners: collection

Admin: daily operation

Timeline is a proposal for discussion with the developer

WHY OUR TEAM SHOULD WIN

CircularFit turns the handover renovation rush into a measurable circular-economy programme. Instead of another marketplace, it routes every strip-out item up the waste hierarchy, to a neighbour, a salvage partner or a recycler, before landfill. Residents log and agree in minutes. Township management gets a handover planner, a routing console and an auditable ledger of tonnes, skip bins and carbon avoided. It is built on Sime Darby Property's own handover moment, scales to any Malaysian township, and our working prototype already runs the full loop.

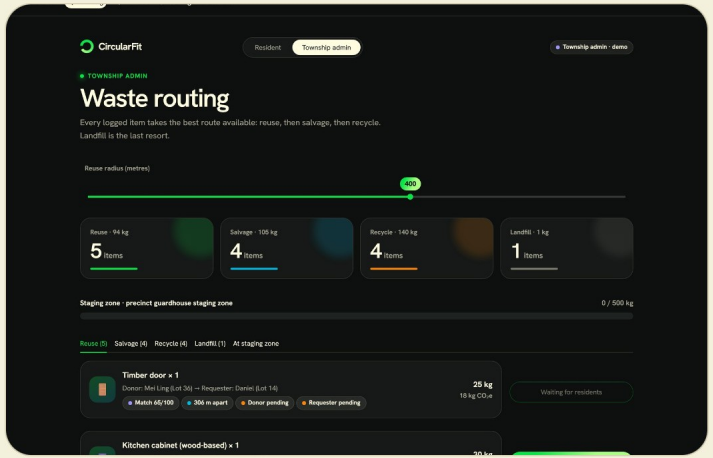


Thank you

[Team name]
[Email · phone]

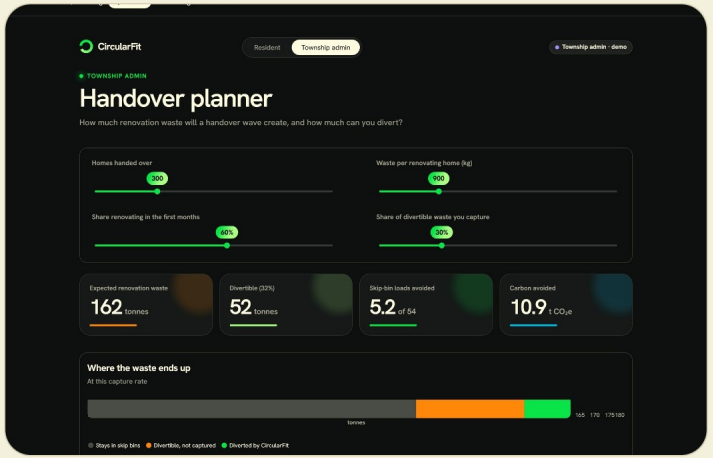
APPENDIX A1 · VISUAL CONCEPT

Inside the prototype



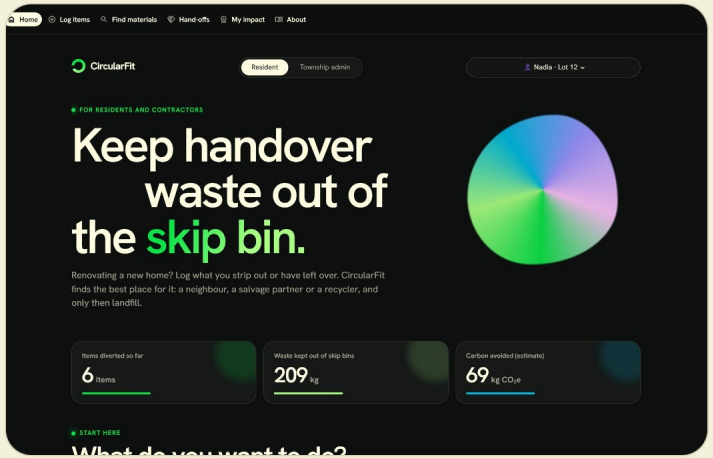
Routing console

Admin sees both residents' answers before confirming.



Handover planner

Forecasts waste, skip-bin loads and carbon for a precinct.



Resident home

Three clear next steps and the township's running totals.

Screens from the working prototype (synthetic data)

APPENDIX A2 · METHOD

How carbon avoided is estimated

carbon avoided = weight × emission factor × condition credit × route credit

Material	kg per unit	kg CO_2e per kg	Source
Ceramic floor tile (box)	20	0.80	ICE v4 factor, weight assumed
Porcelain tile (box)	22	0.76	ICE v4 factor, weight assumed
Sink, basin or WC (ceramic)	15	1.93	ICE v4 factor, weight assumed
Timber door	25	0.70	Placeholder, to verify
Kitchen cabinet (wood-based)	30	0.90	Placeholder, to verify
Tap or faucet (brass)	1.2	3.00	Placeholder, to verify

Condition credit: new 100%, good 80%, fair 50%. Route credit: reuse 100%, salvage 80%, recycle 15%, landfill 0%. Production stage (A1–A3) only; transport and end-of-life are ignored.

Rows marked placeholder are guesses to be replaced with cited values

APPENDIX A3 · ROLES

Two roles, one agreement flow

1
Match proposed
The system ranks donor and requester pairs

2
Both agree
Each resident says yes or no on their own page

3
Admin confirms
A second pop-up with a tick box

4
Ledger and points
Only confirmed hand-offs count

A decline by either resident sends the item back to routing.

Resident
Log items, find materials, agree or decline hand-offs, see points

Township admin
Routing, confirm hand-offs, planner, settings, dashboard

Roles are a demo switch in the prototype; a live version needs real sign-in

APPENDIX A4 · DATA AND LIMITS

What is real, and what is assumed

Sources

- Ain Shams Engineering Journal (2021): housing-sector waste and its reusable share
- UTHM study: construction waste and landfill space in Malaysia
- ICE database v4, via Climatiq: tile and sanitaryware emission factors

Limits

- All residents, listings and hand-offs are synthetic
- Materials are picked by the user; no vision model yet
- Demo sign-in only; no real accounts or live backend
- Route credits, staging size, recycler minimum and skip-bin size are assumptions
- Used fixtures need a condition check and a no-warranty notice

All data sources and assumptions are stated, as the hackathon requires